

Project Report: HR Analytics Prediction and Visualization

Abstract

This project focuses on predicting employee attrition using machine learning and visualizing HR insights through Power BI. The main objective is to help organizations understand the key factors that contribute to employee turnover and identify high-risk employees. By combining data-driven prediction with interactive visualization, the project provides a comprehensive view of workforce dynamics to improve retention strategies.

Introduction

- Employee attrition poses a significant challenge for human resource departments, impacting productivity, hiring costs, and employee morale.
- This project applies data science techniques to predict which employees are likely to leave and visualizes the results using Power BI dashboards for better decision-making.
- The model analyzes attributes such as age, salary, job role, performance rating, and overtime to understand their influence on attrition trends.

Tools and Technologies Used

- **Python (Pandas, NumPy, Scikit-learn):** For data preprocessing, model building, and evaluation.
- **Power BI:** For creating interactive dashboards and visual analytics.
- **Jupyter Notebook:** To run and document the machine learning workflow.
- **SHAP Library:** To interpret and explain feature importance.
- **CSV Files:** For exchanging data between Python and Power BI.

Steps Involved in Building the Project

1. Data Collection and Preparation

- Imported HR dataset containing details such as employee demographics, work history, and attrition status.
- Handled missing values and converted categorical variables using **Label Encoding**.
- Split the data into training and testing sets for model evaluation.

2. Model Building and Training

- Used Decision Tree Classifier from Scikit-learn to predict employee attrition.
- Evaluated the model using accuracy, precision, recall, and F1-score metrics.
- Identified key influencing factors using SHAP feature importance.

3. Exporting Predictions

- Predicted attrition probabilities for test employees and saved the results in a `hr_predictions.csv` file.
- The exported file was then used as a data source for visualization in Power BI.

4. Power BI Dashboard Visualization

- Designed an interactive HR Analytics Dashboard showing:
 - Attrition Rate by Department and Gender
 - Salary vs Attrition
 - Overtime Impact on Attrition
 - Key KPIs: Total Employees, Active Employees, Attrition Percentage
- Added slicers for department, gender, and education to allow user-driven analysis.

Conclusion

The HR Analytics Prediction project effectively integrates machine learning and data visualization to deliver actionable insights. The predictive model identifies employees at risk of leaving, while the Power BI dashboard enables HR managers to visualize and interpret attrition trends in real time. This combined approach enhances workforce planning and supports evidence-based HR decision-making.